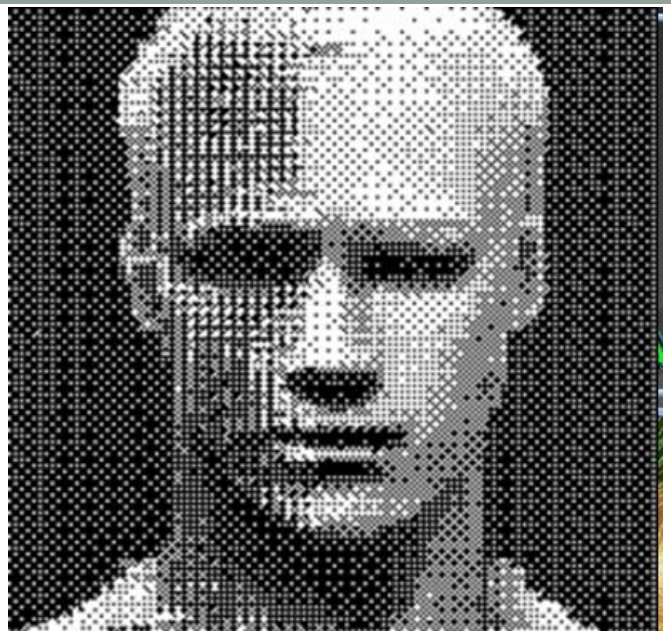


DIGITAL IMAGE PROCESSING

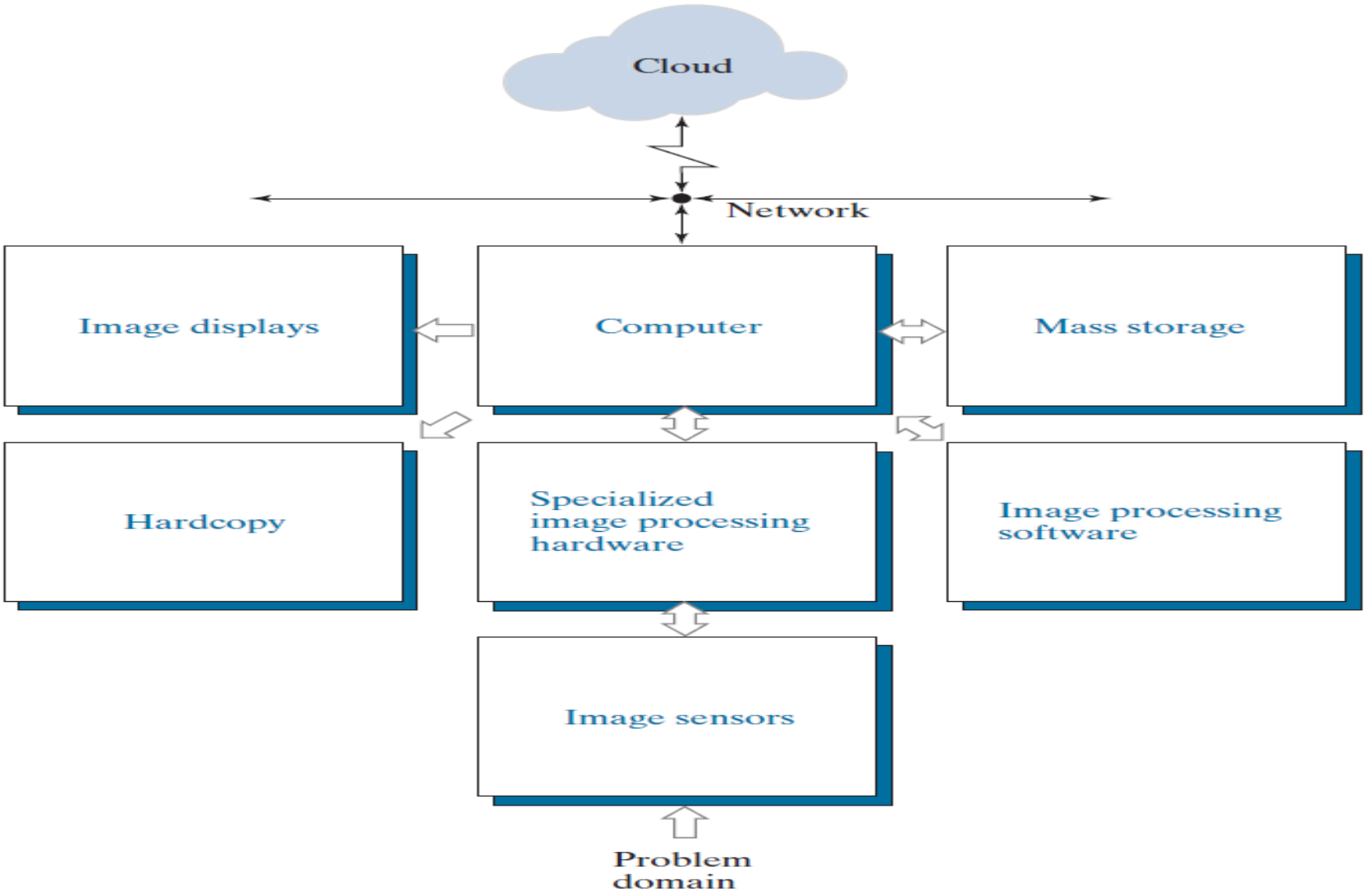
Dr. Manar Donia

CMPn332 - CMPN332 & CMP 432

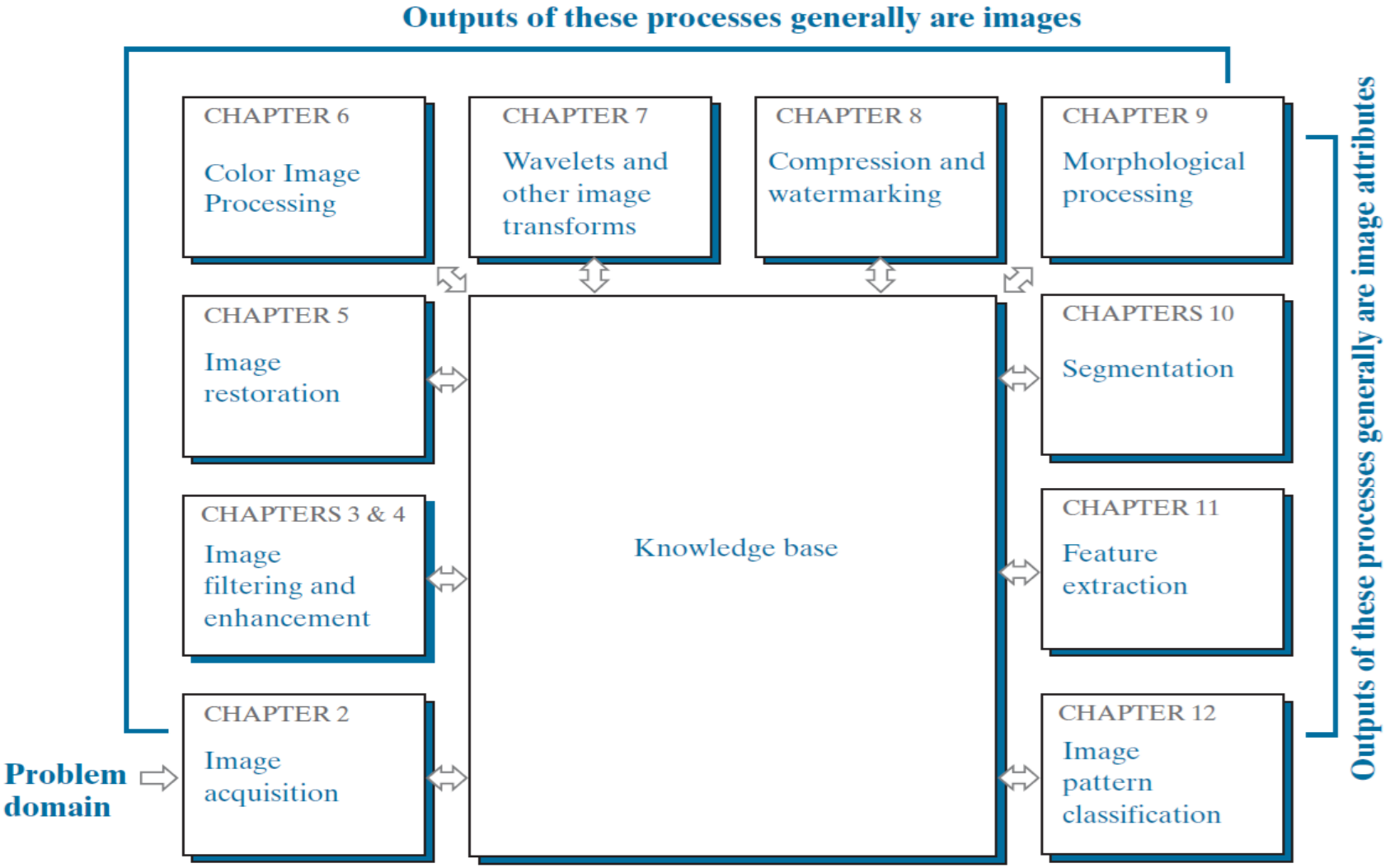
Revision



Basic components of any DIP system



Fundamental steps in digital image processing



Relationships between pixels

- Pixel relationship helps define the structure, content, and the image's visual quality.
- They are vital in many tasks such as image enhancement, segmentation, edge detection, and feature extraction.
- key types of relationships between pixels:
  - Neighboring
  - Connectivity (adjacent)
  - Labelling (connected regions)
  - Distance measures

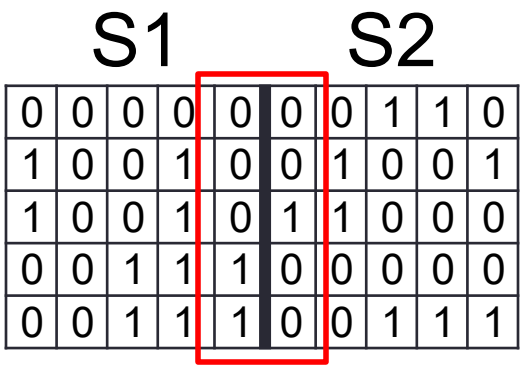
Connectivity (adjacent)

- Two pixels are said to be connected(adjacent) if they are neighbors, and their gray levels satisfy a similarity (belong to the same connectivity set).

Consider the two image subsets, S1 and S2, shown in the following figure. For connectivity set = {1}, determine whether these two subsets are

a. 4 adjacent

b. 8 adjacent



Answer

a. The two subsets is not 4 adjacent, its not possible

b. The two subsets is 8 adjacent

Consider the image segment shown Let connectivity set= {0,1} Compute the following

a. N4, Nd ,N8 for pixel (1,1)
b. Shortest distance between p and q
c. D4 between p and q
d. D8 between p and q
e. Dm between p and q

P(3,0)

3	1	2	3
2	2	3	2
1	2	1	1
1	0	1	2

q(0,3)

0	1	0	0
0	0	0	0
2	0	3	3
2	2	3	0

L1=P(0,1)
L2=L3=p(2,0), p(3,0),P(2,2),p(2,3),p(3,2)

Answer

a. $N_4(f(1,1)) = \{f(0,1), f(2,1), f(1,0), f(1,2)\} = \{1, 2, 2, 3\}$

b. $N_d(f(1,1)) = \{f(0,0), f(0,2), f(2,0), f(2,2)\} = \{3, 2, 1, 1\}$

c. $N_8(f(1,1)) = \{f(0,1), f(2,1), f(1,0), f(1,2), (0,0), f(0,2), f(2,0), f(2,2)\}$

$$N_8(f(1,1)) = \{1, 2, 2, 3, 3, 2, 1, 1\}$$

a. Shortest distance Euclidian distance= $D_e(\mathrm{p},\mathrm{q}) = \sqrt{(x-s)^2 + (y-t)^2} = 4.2$

b. D_4(City Block distance) between p and q= $|x-s| + |y-t| = 6$

c. D_8(Chess board distance) between p and q= $\max(|x-s|, |y-t|) = 3$

d. Dm between p and q depend on connectivity set= {0,1} = not possible

Example: the image shown in the following image matrix find the histogram?

3	1	2	1
2	2	3	2
1	2	1	1
1	0	1	2

Answer

Gray level	frequency
0	1
1	7
2	6
3	2

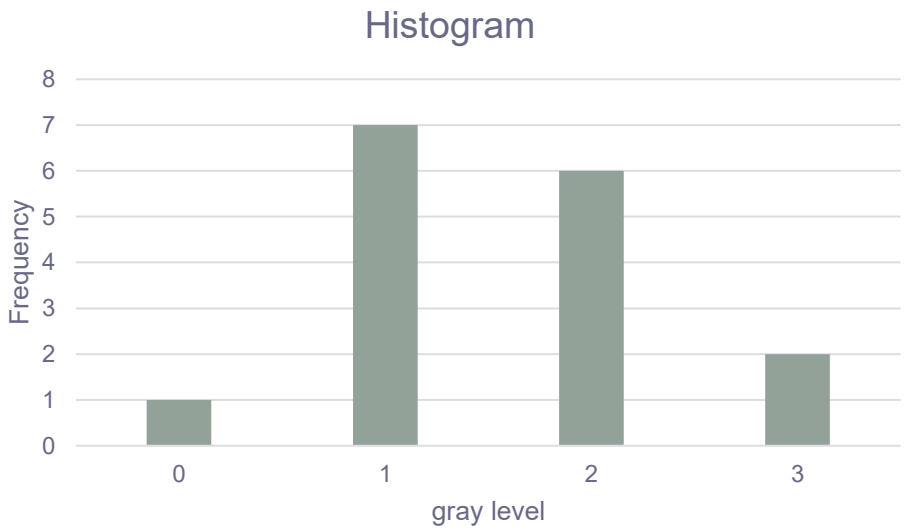


Image Enhancement

Image enhancement is done by combining geometric transformation and enhancement techniques to improve the visual quality of images using **Cascading Operations**.

Geometric transformation change the spatial arrangement of pixels to correct geometric distortions that occur with non-ideal camera angles.

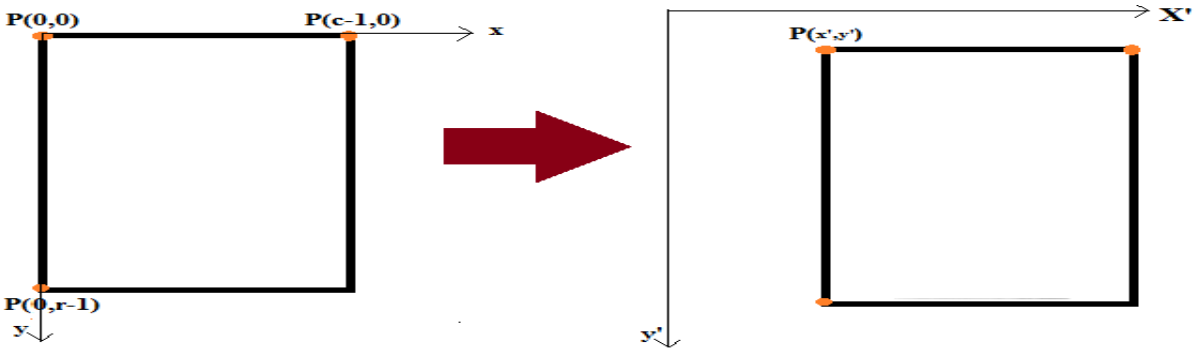
While enhancement techniques modify pixel values without changing their spatial relationships.

Example:

- **Geometric Transform:** Scale and rotate the image to a standard size and orientation.
- **Enhance**: Apply enhancement algorithm for noise reduction and contrast adjustment.

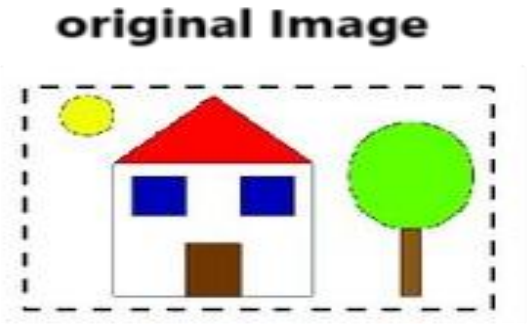
Geometric Transformation

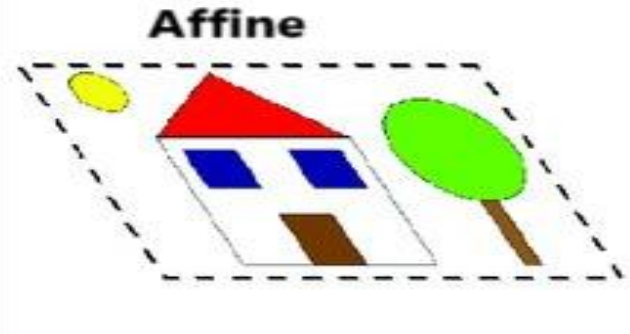
Geometric transformations modify the image's geometry, allowing position, orientation, size, and shape **without altering the underlying pixel values**. **f***(x',y')=**f**(x,y).*



Geometric transformations can be categorized into they differ primarily in how th**Affine transformations** and **non-affine transformations**,ey manipulate the geometry of images.

Geometric Transformation







Aspect	Affine Transformations	Non-Affine Transformations
Definition	Preserve points, straight lines, and planes.	Do not preserve lines; may warp or bend the image.
Linear vs. Non-Linear	Linear transformations.	Non-linear transformations.
Parallelism	Maintains parallelism (parallel lines remain parallel).	Does not maintain parallelism.
Distance Ratios	Preserves ratios of distances between points.	Does not preserve distance ratios.
Transformation	Can be expressed with a matrix in a simple form.	Often requires more complex functions for representation.
Examples	Rotating an image by 30 degrees, scaling an image to half its size.	Creating a 3D perspective effect, bending an image.

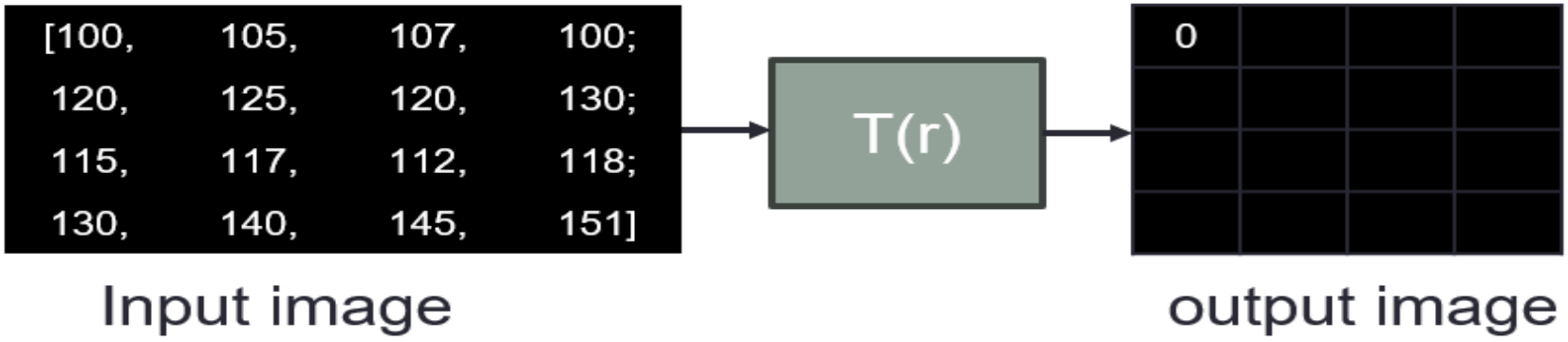
Image enhancement Point processing

The simplest formula for image enhancement technique is:

$$S=T(r)$$

Where:

S: pixel value of output image after processing
T: transformation that maps image pixel r into
r: pixel value of input image before processing



Key point processing techniques used in image enhancement are:

- **Histogram Equalization:**
  It is a simple way to improve dark or washed-out images (low-dynamic range) by Redistributes pixel intensities across the entire range of available values based on CDF.

  $$S = CDF(r) \times (L-1)$$

- **Contrast Stretching:**
  preferred for enhancing images with a known intensity range needed to be enhanced.

  $$\alpha = \frac{s_1 - 0}{r_1 - 0}, \beta = \frac{s_2 - s_1}{r_2 - r_1}, \gamma = \frac{L - 1 - s_2}{L - 1 - r_2}$$

  $$s = \begin{cases} \alpha * r & 0 \le \boldsymbol{r} < r_1 \\ \beta\,(r - r_1) + s_1 & r_1 \le \boldsymbol{r} < r_2 \\ \gamma(r - r_2) + s_2 & r_2 \le \boldsymbol{r} < L - 1 \end{cases}$$

- **Image Negative**
  useful for enhancing white or grey detail embedded in dark regions of an image

  $$S = (L-1) - r$$

- **Logarithmic transformation**
  Enhancing image details in darker regions while compressing lighter areas

  $$S = c * \log(1 + r) \qquad \text{where } c = (L - 1)/\log(L)$$

- **Gamma Correction (power law)**
  depend on the value of γ
  - γ <1 map a narrow range of dark input values into a wider range of output values.
  - γ >1 map a narrow range of bright input values into a wider range of output values.

  $$s = c * r^{\gamma}$$

- **Gray level slicing**
  It's a way to highlight gray range of interest to a viewer by one of two ways:
  1. Transformation highlights range [A, B] of gray level and reduces all others to a constant level

     $$S(x,y)\begin{cases} 255 & if\ \ A \le r(x,y) \le B \\ C & otherwise \end{cases}$$

  2. Transformation highlights range [A, B] but preserves all other levels

     $$S(x,y)\begin{cases} 255 & if\ \ A \le r(x,y) \le B \\ r(x,y) & otherwise \end{cases}$$

- **Bit plane slicing** represents an image with one or more bits of the byte used for each pixel.
  The main goals of bit plane slicing are:
  - Converting a gray-level image to a binary image.
  - Image Compression (Representing an image with fewer bits) where the most significant data is represented in MSB

Histogram Equalization

A 5-bit 40x40 image has the following intensities, Find out the equalized histogram gray levels for the image.

Gray level	Frequency
11	580
15	320
10	150
20	250
18	300

- **Answer**:
- A 5-bit →$L=2^5 = 32$
- 40x40 → M*N=40*40=1600

Gray level r_k	Frequency	Probability= $P_r(r_k)=\frac{count}{M*N}$	CDF=$\sum_{j=0}^{k} p_r(rj)$	Equalized grays= $(L-1)*CDF$
10	150	0.093	0.093	3
11	580	0.362	0.456	14
15	320	0.2	0.656	20
18	300	0.187	0.843	26
20	250	0.156	1	31

The histogram equalization stretch out the intensity range from 10:20 to become 3:31 which improve the image contrast

Contrast stretching

- Example:
- A 3-bit 4x4 image has the following intensities, Find out the output gray levels after applying contrast stretching. Using r1=3, r2=5 s1=2 and s2=6

Answer:

[4,	3,	5,	2;
3,	6,	4,	6;
2,	2,	6,	5;
7,	6,	4,	1]

$$\alpha = \frac{s_1 - 0}{r_1 - 0} = 0.66,\ \beta = \frac{s_2 - s_1}{r_2 - r_1} = 2,\ \Upsilon = \frac{7 - s_2}{7 - r_2} = 0.5$$

$$s = \begin{cases} \alpha * r & 0 \le \boldsymbol{r} < r_1 \\ \beta\,(r - r_1) + s_1 & r_1 \le \boldsymbol{r} < r_2 \\ \Upsilon(r - r_2) + s_2 & r_2 \le \boldsymbol{r} < L - 1 \end{cases}$$

$$s = \begin{cases} 0.66 * r & 0 \le \boldsymbol{r} < 3 \\ 2 * (r - 3) + 2 & 3 \le \boldsymbol{r} < 5 \\ 0.5 * (r - 5) + 6 & 5 \le \boldsymbol{r} < 7 \end{cases}$$

[4,	2,	6,	1;
2,	7,	4,	7;
1,	1,	7,	6;
7,	7,	4,	1]

Point processing

Giving the following image F(x,y)

120	55	43	123
14	12	89	78
231	34	129	65
91	200	157	57

- Which image enhancement technique used to enhance white or grey detail embedded in dark regions of an image, show enhancement effect at pixel (3,1).
- Which image enhancement technique used to enhance image details in darker regions while compressing lighter areas, show enhancement effect at pixel (2,0).
- Which image enhancement technique used to enhance image with high brightness , show enhancement effect at pixel (2,3).
- Which image enhancement technique used to highlight gray range of interest to a viewer, show enhancement effect at pixel (3,0) for Range 20 to 80.
- Bit plane slicing is a method of representing an image with different planes which plane represent the most significant data. Show its content for pixel (1,1)

- Image negative→S= (L-1) – r
- Log transformation→ S = c * log (1 + r)
  where c= (L - 1) / log(L)
- Power law→$\boldsymbol{S = c * r^{\gamma}}$
- Gray Level slicing

$$S(x,y)\begin{cases}255 & if\ A < r(x,y) < B \\ C & otherwise\end{cases}$$

$$S(x,y)\begin{cases}255 & if\ A < r(x,y) < B \\ r(x,y) & otherwise\end{cases}$$


Thank you!